NAKAJIMA'S REPRESENTATION OF THE HEISENBERG ALGEBRA

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Abstract

The main theorem in [Nak95] shows there is an action of the infinite-dimensional Heisenberg algebra on the homology groups of the Hilbert scheme of points on a projective surface. We explain how this representation is constructed and give an idea of the proof of the theorem.

Contents

1. Introduction

Motivated by earlier work on moduli spaces of torsion-free sheaves (see [Nak94]), Nakajima constructed in [Nak95] a representation the infinite-dimensional Heisenberg algebra on the homology groups of the Hilbert schemes on a nonsingular projective surface.

In this work we outline the basic ideas in Nakajima's construction and explain the first part of the proof of the main theorem. Most of this work is based in [Nak], with the exception of the first section where I try to identify the precise version of the Heisenberg algebra that is being represented. In Section 3, I explain the construction of the Hilbert scheme, providing slightly more details than what is really essential to understand the main theorem, but which help to get a better idea on what the Hilbert scheme is and how to work with it.

This work is strongly influenced by [vid21], a series of seminars aimed at understanding [Nak] documented in videos. This was of great help specially in Section 4, since the properties of Borel-Moore homology are a rather technical but essential part of the construction. The remaining sections, containing the main theorem and part of its proof, also pick up ideas from these videos, but are mostly based in [Nak].

2. Heisenberg algebra

In this section we review four different variants of the Heisenberg algebra and discuss their characters. We conclude by showing how the character is used to prove that Nakajima's representation is of highest weight.

The most basic definition of the Heisenberg Lie algebra is $\hat{\mathfrak{a}} = \bigoplus_{n\in\mathbb{Z}} \mathbb{C}h_n \oplus \mathbb{C}K$ with bracket given by

$$[h_m, h_n] = m\delta_{m,-n}K, \qquad [K, \hat{\mathfrak{a}}] = 0. \tag{2.0.1}$$

In class we described its highest weight representation $H = \mathbb{C}[x_1, x_2, \dots]$ with

$$h_n \mapsto \begin{cases} n\frac{\partial}{\partial x_n} & \text{if } n > 0 \\ x_{-n} & \text{if } n < 0 \\ 0 & \text{if } n = 0 \end{cases}, \qquad K \mapsto \mathrm{Id}.$$

We introduced a grading in $H = \bigoplus_{\Delta=0}^{\infty} H_\Delta$ given by the *energy* of monomials, defined by

$$\Delta(x_1^{k_1} x_2^{k_2} \dots x_s^{k_s}) = \sum_j k_j,$$

which gives the character formula

$$\sum_{\Delta=0}^{\infty} q^{\Delta} \dim H_\Delta = \prod_{m=1}^{\infty} \frac{1}{1-q^m}. \tag{2.0.2}$$

In order to generalize this construction it's worth going over why this formula holds. Recall that the *Verma module* of weight $\mu = 0$ of $\hat{\mathfrak{a}}$ is defined by $H = \mathcal{U}(\hat{\mathfrak{a}}) \otimes_{\mathcal{U}(\hat{\mathfrak{a}}_+)} \mathbb{C}|0\rangle$. Further, by the PBW theorem, this module is isomorphic to $\mathcal{U}(\mathfrak{n}_-)$, the universal enveloping algebra of the negative roots in the triangular decomposition (as in the finite-dimensional case, see [Kac10, Lecture 24] and [vid21, Video 8]). Since the subalgebra of negative roots is abelian, the universal enveloping algebra becomes the symmetric algebra and therefore we have

$$\begin{aligned} H &\cong \mathrm{Sym}\left(\bigoplus_{m>0} \mathbb{C}at^{-m}\right) \\ &\cong \bigotimes_{m>0} \mathrm{Sym}(\mathbb{C}at^{-m}) \\ &\cong \bigotimes_{m>0} \mathbb{C}[at^{-m}], \end{aligned} \tag{2.0.3}$$

since the symmetric power of a vector space is the polynomial ring in its generators, and the symmetric power of a direct sum is the tensor product of the symmetric powers of the factors. This matches our previous formula $H = \mathbb{C}[x_1, x_2, \dots]$ by letting $x_m = at^{-m}$.

Putting a dummy variably q to every 1-dimensional factor and multiplying throughout, we obtain

$$\begin{aligned} m=1 &\qquad (1+q+q^2+q^3+\dots)\times \\ m=2 &\qquad (1+q^2+q^4+q^6+\dots)\times \\ m=3 &\qquad (1+q^3+q^6+q^9+\dots)\times\dots, \end{aligned}$$

so that indeed the character is given by Equation 2.0.2.

Another version of this construction we discussed in lectures (see [vE23, Section 3.1]) starts with any finite-dimensional vector space $\mathfrak{h}$ equipped with a symmetric bilinear form $(\cdot,\cdot)$. Define

$$\hat{\mathfrak{h}} = \mathfrak{h}\otimes\mathbb{C}[[t^{\pm1}]]\oplus\mathbb{C}K$$

with bracket

$$[at^m, bt^n] = m\delta_{m,-n}(a,b)K, \qquad [K,\hat{\mathfrak{h}}]=0, \tag{2.0.4}$$

which is just the affine Lie algebra of $\mathfrak{h}$ when regarded as a commutative Lie algebra.

A similar formula to Equation 2.0.3 is

$$\begin{aligned} H &\cong \operatorname{Sym}\left(\bigoplus_{m>0}\bigoplus_{a\in B}\mathbb{C}at^{-n}\right) \\ &\cong \bigotimes_{m>0}\bigotimes_{a\in B}\operatorname{Sym}(\mathbb{C}at^{-m}) \\ &\cong \bigotimes_{m>0}\bigotimes_{a\in B}\mathbb{C}[at^{-m}] \end{aligned} \tag{2.0.5}$$

where B is a basis of the underlying vector space $\mathfrak{h}$. If $\mathfrak{h}$ has dimension r, this has character

$$\prod_{m=0}^{\infty}\frac{1}{(1-q^m)^r}. \tag{2.0.6}$$

Taking this construction one step further, let's change $\mathfrak{h}$ for a finite-dimensional super vector space $\mathfrak{h}=\mathfrak{h}_0\oplus\mathfrak{h}_1$ with basis $B^{\text{even}}\amalg B^{\text{odd}}$. If we equip $\mathfrak{h}$ with a graded commutative pairing $\langle\cdot,\cdot\rangle$, we can apply the same construction as above to obtain the algebra

$$\hat{\mathfrak{h}} = (\mathfrak{h}_0[[t^{\pm1}]]\oplus\mathfrak{h}_1[[t^{\pm1}]])\oplus\mathbb{C}K \tag{2.0.7}$$

with Lie bracket

$$[at^m, bt^n] = \langle a,b\rangle\, m\delta_{m,-n}K, \qquad [K,\hat{\mathfrak{h}}]=0. \tag{2.0.8}$$

One can check that this is indeed a super Lie bracket (which essentially holds by the pairing being graded commutative and K being central), i.e. $\hat{\mathfrak{h}}$ is a Lie superalgebra. Its highest weight module is given by

$$\begin{aligned} H &\cong \operatorname{Sym}\left(\bigoplus_{m>0}\bigoplus_{a\in B}\mathbb{C}at^{-m}\right) \\ &\cong \bigotimes_{m>0}\operatorname{Sym}\left(\bigoplus_{a\in B^{\text{even}}}\mathbb{C}at^{-m}\right)\otimes\bigotimes_{m>0}\Lambda\left(\bigoplus_{a\in B^{\text{odd}}}\mathbb{C}at^{-m}\right). \end{aligned}$$

since the universal enveloping algebra of the odd part is the exterior power (see [vid21, Video 8]).

Now recall that while the symmetric power of a vector space yields the polynomial algebra on the basis of the vector space, the exterior power yields the algebra generated by the wedge products of the basic vectors. In the case of a 1-dimensional vector space generated by the element at^{-m}, we get

$$\Lambda(\mathbb{C}at^{-m}) = \mathbb{C} \oplus \mathbb{C}at^{-m}$$

and thus the character contribution in this case is simply $1 + q^m$. Multiplying throughout we obtain the character

$$\prod_{m=0}^{\infty} \frac{(1+q^m)^{\dim \mathfrak{h}^{\mathrm{odd}}}}{(1-q^m)^{\dim \mathfrak{h}^{\mathrm{even}}}}.$$

Let's introduce one last variant of this construction. This time we start from a graded vector space $\mathfrak{h} = \bigoplus_{m\in\mathbb{Z}} \mathfrak{h}_m$, which is automatically also a supervector space by the $\mathbb{Z}_2$-grading $\mathfrak{h}^{\mathrm{even}} = \bigoplus_{i\in\mathbb{Z}} \mathfrak{h}_{2i}$ and $\mathfrak{h}^{\mathrm{odd}} = \bigoplus_{i\in\mathbb{Z}} \mathfrak{h}_{2i+1}$. Suppose $\mathfrak{h}_i$ is finite-dimensional with basis B_i, and that $\mathfrak{h}$ is equipped with a graded commutative pairing $\langle\cdot,\cdot\rangle$. The definition of $\hat{\mathfrak{h}}$ as an algebra is given exactly as in Equation 2.0.7 with bracket given by Equation 2.0.8.

Now we obtain

$$\begin{aligned} H &\cong \mathrm{Sym}\left(\bigoplus_{m>0} \bigoplus_{\substack{a\in B_i\\ i\in\mathbb{Z}}} \mathbb{C}at^{-m}\right) \\ &\cong \bigotimes_{m>0} \mathrm{Sym}\left(\bigoplus_{\substack{a\in B_{2i}\\ i\in\mathbb{Z}}} \mathbb{C}at^{-m}\right) \otimes \bigotimes_{m>0} \Lambda\left(\bigoplus_{\substack{a\in B_{2i+1}\\ i\in\mathbb{Z}}} \mathbb{C}at^{-m}\right). \end{aligned}$$

This gives the character

$$\sum_{\substack{\Delta\in\mathbb{Z}_{\geq 0}\\ i\in\mathbb{Z}}} \dim \mathfrak{h}_{i,\Delta} t^i q^\Delta = \prod_{m>0}\prod_j (1-(-1)^j t^j q^m)^{-(-1)^j \dim \mathfrak{h}_j}.$$

Finally fix $\mathfrak{h} = H_*(X)$, the singular homology ring of a complex projective surface X. Then the character of the corresponding Heisenberg algebra is

$$\prod_{m=1}^{\infty} \frac{(1+t^{2m-1}1^m)^{b_1(X)}(1+t^{2m+1}q^m)^{b_3(X)}}{(1-t^{2m-2}q^m)^{b_0(X)}(1-t^{2m}q^m)^{b_2(X)}(1-t^{2m+2}q^m)^{b_4(X)}}. \tag{2.0.9}$$

It was proved by Göttsche using the Weil conjectures in [Gö90] (and in [Nak, Section 6.2] by other methods) that Equation 2.0.9 in fact equals

$$\sum_{n=0}^{\infty} q^n P_t(X^{[n]}),$$

where the *Poincaré polynomials* $P_t(Z)$ are defined by $P_t(Z) = \sum_{m\geq 0} b_m(Z)t^m$, $b_m(Z) = \mathrm{rank} H_m(Z)$ for any variety Z.

The representation of the Heisenberg algebra we discuss in this work corresponds precisely to this version of the Heisenberg algebra. Göttsche's formula shows that it is a highest weight representation, as explained in [Nak, Corollary 8.16].

3. Hilbert scheme of points

In this section we define of the Hilbert scheme of points on a smooth projective complex surface X. The essential property of the Hilbert scheme is that its points (over $\mathbb{C}$) are 0-dimensional closed subschemes of X of length n.

Consider a contravariant functor

$$\mathcal{H}ilb_X : (\text{Schemes})^{\text{op}} \to (\text{Sets})$$

which assigns to any scheme U the set of flat families of closed subschemes in X parametrized by U. Any such family is by definition a flat morphism

$$\begin{tikzcd} Z \arrow[d, "\pi"] & \subseteq X \times U \\ U. & \end{tikzcd}$$

We define $\mathcal{H}ilb$ to map a morphism of schemes $U \to U'$ to the map of sets (of families over U and U') given by pullback. That is, we map a flat family $Z' \to U'$ to $Z' \times_{U'} U \to U$, which remains flat.

The Hilbert polynomial at a fibre Z_u is defined by $P_u(m) = \chi(\mathcal{O}_{Z_u} \otimes \mathcal{O}_X(m))$. By flatness of π the Hilbert polynomial is constant along U ([Har77, Part III, Theorem 9.9]). For a fixed polynomial P we define $\mathcal{H}ilb_X^P$ as the functor which maps U to the set of families over U with Hilbert polynomial P. A theorem by Grothendieck says that $\mathcal{H}ilb_X^P$ is representable by a projective scheme Hilb_X^P. This means that there is a natural transformation $\mathrm{Hom}(-, \mathrm{Hilb}_X^P) \simeq \mathcal{H}ilb_X^P(-)$; that is, families over any scheme U are in bijection with morphisms $U \to \mathrm{Hilb}_X^P$.

As with any representable functor, we obtain a distinguished element associated to the identity morphism of the object representing the functor. In our case, this is a "universal" family $\mathcal{Z} \to \mathrm{Hilb}_X^P$ with the property that any other family $Z \to U$ is obtained as a pullback of $\mathcal{Z}$. More exactly, when regarding an arbitrary family $Z \to U$ as a morphism $f \in \mathrm{Hom}(U, \mathrm{Hilb}_X^P)$, we see that

$$\begin{tikzcd} \mathrm{id}_{\mathrm{Hilb}_X^P} \arrow[d, mapsto] & \mathrm{Hom}(\mathrm{Hilb}_X^P, \mathrm{Hilb}_X^P) \arrow[r, "\simeq"] \arrow[d, "f^*"'] & \mathcal{H}ilb_X^P(\mathrm{Hilb}_X^P) \arrow[d, "U \times_{\mathrm{Hilb}_X^P} -"] & \mathcal{Z} \arrow[d, mapsto] \\ f & \mathrm{Hom}(U, \mathrm{Hilb}_X^P) \arrow[r, "\simeq"] & \mathcal{H}ilb_X^P(U) & Z = U \times_{\mathrm{Hilb}_X^P} \mathcal{Z}. \end{tikzcd}$$

commutes by naturality. This shows that studying families of subschemes of X parametrized by any scheme U can be done by studying the universal family over the Hilbert scheme.

We can formally view subschemes of X as points of Hilb_X^P as follows. Recall that, as with any other scheme, points h of Hilb_X^P correspond to morphisms $\mathrm{Spec}(\kappa(h)) \to \mathrm{Hilb}_X^P$, where $\kappa(h)$ is the residue field of the point (see Stacks Project 01J9). Then for every point h of the Hilbert scheme we have a unique subscheme of X defined as the fibre X_h of the corresponding morphism,

$$\begin{tikzcd} X_h \arrow[r] \arrow[d] \arrow[dr, phantom, "\lrcorner", very near start] & \mathcal{Z} \arrow[d] & \subset X \times \mathrm{Hilb}_X^P \\ \mathrm{Spec}(\kappa(h)) \arrow[r] & \mathrm{Hilb}_X^P. \end{tikzcd}$$

As a rather technical observation that will not be addressed later, we note that to make sure that the fibres are $\mathbb{C}$-schemes, we must restrict our analysis to $\mathbb{C}$-points of the Hilbert scheme (i.e. when the residue field is $\mathbb{C}$).

Definition 3.1. Let P be a constant polynomial given by $P(m) = n$ for all $m \in \mathbb{Z}$. The *Hilbert scheme of n points in X* is $X^{[n]} := \mathrm{Hilb}_X^P$.

This condition forces the dimension of the fibres of any family over X to be zero. Indeed, the leading term of the Hilbert polynomial of a projective variety of degree d and dimension s is $(dm^s)/s!$, which can only be constant for $s = 0$ (see [HM98, Exercise 1.13]).

Further, any 0-dimensional subscheme of X is supported on finitely many points of X (this is consequence of X being Noetherian: we couldn't have infinitely many points since they would give an infinite chain of prime ideals). Finally, the *length* of any such subscheme, defined by $\dim \Gamma(\mathcal{O}_Z)$ must be $\sum_{x \in \mathrm{supp} Z} \dim \Gamma(\mathcal{O}_{Z_x}) = n$. (see Stacks Project 0AYT).

It is now easy to see that any subscheme of X consisting of n distinct points is itself a point of Hilb_X^P. Indeed, if $Z = \{x_1, \ldots, x_n\} \subset X$, its structure sheaf is the direct sum of skyscraper sheaves

$$\mathcal{O}_Z = \bigoplus_{i=1}^{n} (\text{skyscraper sheaf at } x_i)$$

so that $\mathcal{O}_Z \otimes \mathcal{O}_X(m) \cong \mathcal{O}_Z$ and thus the Hilbert polynomial of Z is constant n. The latter isomorphism can be verified via the exact sequence associated to the maximal ideal $\mathfrak{m}_{x_i}$ of the stalk $\mathcal{O}_{X,x_i}$, namely

$$0 \longrightarrow \mathfrak{m}_{x_i} \longrightarrow \mathcal{O}_{X,x_i} \longrightarrow \mathcal{O}_{X,x_i}/\mathfrak{m}_{x_i} \cong \mathbb{C} \longrightarrow 0,$$

which after tensoring by $\mathcal{O}_{Z,x_i} \cong \mathbb{C} \cong \mathcal{O}_{X,x_i}/\mathfrak{m}_{x_i}$ kills the first term. Twisting by m does not alter this construction.

However, not every point in $X^{[n]}$ is given by a set of n distinct points, nor by a set of points with multiplicites, as the following construction suggests. Define the *symmetric power* of X as

$$S^n X = \underbrace{X \times \ldots \times X}_{n \text{ copies}} / S_n$$

where the symmetric group S_n acts by permutating copies. Thus, a point of $S^n X$ is an S_n-orbit on the cartesian product $X \times \ldots \times X$, i.e., an unordered n-tuple of points of X with allowed repetition. In algebraic geometry these are called "algebraic cycles" and are denoted as formal sums $\sum_{x \in X} n_x[x]$ where $\sum_{x \in X} n_x = n$, with $n_x \in \mathbb{N}$, and $[x]$ are formal symbols.

By our discussion above on the length of 0-dimensional subschemes of X, we have a map

$$\begin{aligned}\pi : X^{[n]} &\longrightarrow S^n X \\ Z &\longmapsto \sum_{x \in \operatorname{supp} Z} \dim \Gamma(\mathcal{O}_{Z,x})[x]\end{aligned}$$

called the *Hilbert-Chow morphism.* In some cases this is an isomorphism, but not always.

Theorem 3.2 (Fogarty, [Fog68])**.** *Suppose X is nonsingular and $\dim X = 2$, then the following holds.*

(1) *$X^{[n]}$ is nonsingular and has dimension $2n$.*
(2) *The Hilbert-Chow morphism is a resolution of singularities.*

4. Borel-Moore homology

In this section we define Borel-Moore (BM) homology and formulate its basic properties. Most of this section is based on [vid21, Video 9]. For further details on the construction see [Ful96].

Let M be a a smooth, connected, oriented, finite-dimensional manifold over $\mathbb{R}$. While the following results hold for any topological space $X \subset M$ that is a proper deformation retract of a closed neighbourhood of a manifold, for us it's enough to think of $X \subset M$ as a (possibly singular) variety over $\mathbb{C}$.

In this work we consider homology and cohomology groups with coefficients in $\mathbb{C}$. Recall that singular homology is defined as the homology of the chain complex of finite linear combinations of i-simplices in X. *Borel-Moore homology* $H_\bullet^{BM}(X)$ is the homology of the complex of locally finite linear combinations of singular simplices in X, where "locally finite" means that every point $x \in X$ is in a neighbourhood that intersects only finitely many simplices.

A striking difference between singular and BM homology is that the latter is not homotopy invariant. To see this recall that the homology groups of $\mathbb{R}^2$ are $H_0(\mathbb{R}^2) = \mathbb{C}$ and $H_i(\mathbb{R}^2) = 0$ for $i \neq 0$. However, $H_2^{BM}(\mathbb{R}^2) = \mathbb{C}$ since a tiling of triangles is a locally finite chain of 2 simplices, has no boundary (since the boundaries of all triangles cancel each other), and is clearly not the boundary of a 3-simplex; thus this tiling is a nonzero element of $H_2^{BM}(\mathbb{R}^2)$. Further, notice that $H_0^{BM}(\mathbb{R}^2) = 0$ since a ray of 1-simplices is a locally finite chain whose boundary is a point; thus a point is a boundary and has trivial homology class.

Here are three important properties of BM homology:

(1) If X is compact, $H_i^{BM}(X) \cong H_i(X)$.
(2) If X is not compact, $H_i^{BM}(X) \cong H_i(\hat{X}, \infty)$ where $\hat{X}$ is the one-point compactification of X and $H_i(\hat{X}, \infty)$ is the relative homology group.
(3) If the dimension of the manifold where X is embedded is m, there is a canonical isomorphism $H_i^{BM}(X) \cong H^{m-i}(M, M \setminus X)$. In other words, there is a perfect pairing

$$H_i^{BM}(X) \times H_{m-i}(M, M \setminus X) \to \mathbb{C}$$

given by "intersection" (where the quotes account for some transversal intersection condition that we would require before computing the intersection of two simplicial chains).

Consider the previous example of the tiling of $\mathbb{R}^2$ by triangles and put $X = M = \mathbb{R}^2$. Take any point as a 0-chain in singular homology and the triangle tiling as 2-chain in BM homology. Since the point lies in exactly one triangle (after suitable perturbation) their intersection is 1. Notice this wouldn't work with singular homology: we need an infinite 2-chain to make sure that the whole plane is tiled by triangles since otherwise we cannot guarantee that the point is contained in some triangle.

Now we formulate the key properties of BM homology needed to define the operators in Nakajima's representation of the Heisenberg algebra.

(1) (Pushforward.) If $f : X \to Y$ is proper (i.e. inverse image of any compact set is compact) then there is a pushforward map $f_* : H_i^{BM}(X) \to H_i^{BM}(Y)$. This is functorial in proper maps, i.e. preserves compositions. Roughly, this works because, by properness, the inverse image of a point in Y will be mapped to a compact set of X, which can intersect only finitely many simplices of a given chain, ensuring local finiteness of the image under f of the chain.

(2) (Pullback.) If $f : X \to Y$ is a locally trivial fibration with fibre a manifold of dimension m (over $\mathbb{R}$), then there is a pullback map $f^* : H_i^{BM}(Y) \to H_{i+m}^{BM}(X)$.

(3) (Intersection pairing.) If X and Y are both "nicely embedded" in the same manifold M, there is a bilinear graded-commutative map

$$\cap_M : H_i^{BM}(X) \times H_j^{BM}(Y) \to H_{i+j-m}^{BM}(X \cap Y).$$

Constructing this product can be done via the isomorphism with relative singular cohomology (item (3) in the previous list), and then using usual cap product. Again, the term "nicely embedded" accounts for a transversality condition that we will not specify.

(4) (Fundamental classes.) To any variety V we can associate a fundamental class $[V]$. More precisely,
   - If V is a smooth irreducible variety of dimension d over $\mathbb{C}$, its *fundamental class* $[V]$ is the canonical generator of $H_{2d}^{BM}(V) \cong H^0(V) \cong \mathbb{C}$, where the first isomorphism is given by the isomorphism of BM homology and relative singular cohomology taking $V = M$ since it is smooth.
   - If V is a (possibly singular) irreducible variety of dimension d over $\mathbb{C}$, its fundamental class $[V] \in B_{2d}^{BM}(V)$ is the unique class such that $u^*[V] = [V_{\text{reg}}]$. This is well-defined since the pullback in this degree turns out to be injective (by further properties of BM homology not discussed here) and thus this class is unique.
   - For V not necessarily irreducible of dimension d, then $H_{2d}^{BM}(V)$ has basis $[V_1], \ldots, [V_k]$ where V_i are the d-dimensional irreducible components of V. In this case we define the fundamental class as the sum of the fundamental classes of those components, i.e. $[V] = [V_1] + \ldots + [V_k]$.

As a final remark notice that it is as special feature of BM homology that we can define fundamental classes for noncompact manifolds, as opposed to singular homology.

5. Correspondences

In this section we introduce the main tool for constructing the operators in Nakajima's representation of the Heisenberg algebra.

Let X_1, X_2 be smooth irreducible varieties of dimensions d_1, d_2. Let Z be a d-dimensional closed subvariety of $X_1 \times X_2$ such that the projection onto the first factor is proper.

The *correspondence* associated to $Z \subset X_1 \times X_2$ is the map

$$\begin{aligned} c_{[Z]} : H_i^{BM}(X_2) &\longrightarrow H_{i+2d-2d_2}^{BM}(X_1) \\ c &\longmapsto p_{1,*}(p_2^* c \cap [Z]) \end{aligned} \tag{5.0.1}$$

where p_1 and p_2 are the projections in the following diagram:

$$\begin{tikzcd} & Z \subset X_1 \times X_2 \arrow[dl, "p_1"'] \arrow[dr, "p_2"] & \\ X_1 & & X_2. \end{tikzcd}$$

That is, the correspondence is an operator given by pulling back a class by p_2, intersecting with the fundamental class of $[Z]$, and pushing forward by p_1. This map is well-defined according to the key properties of BM homology stated in the previous section; the degree of the target homology group is computed according to such properties.

Remark 5.1.

(1) We can define correspondences more generally replacing the fundamental class of Z by any $\alpha \in H_{2d}^{BM}(Z)$. In fact, the operators in our representation shall be of this more general type.

(2) If Z is the graph of a proper morphism $f : X_2 \to X_1$, then $c_{[Z]} = f_*$. This shows that correspondences generalize the pushforward in BM homology.

(3) Since X_1 and X_2 are smooth, we may use the isomorphism of BM and relative singular cohomology to identify a class in the tensor product with an endomorphism as we do in general for V, W vector spaces via the usual isomorphism $V \otimes W^* \cong \mathrm{Hom}(W, V)$, $v \otimes \varphi \mapsto (w \mapsto \varphi(w)v)$. In this way the fundamental class $[Z] \in \bigoplus_{i+j=\dim_{\mathbb{R}} Z} H_i(X_1) \otimes H_j^{BM}(X_2)$ immediately gives a map

$$\bigoplus_n H_*(X^{[n]}) \to \bigoplus_n H_*(X^{[n \mp i]})$$

which, in fact, coincides with Equation 5.0.1 (see [Nak, Section 8.3]).

Now we show how to compose correspondences, as we shall do in the proof of the main theorem.

Suppose we are given two subvarieties $Z_{12} \subset X_1 \times X_2$ and $Z_{23} \subset X_2 \times X_3$ such that the projections to X_2 and X_3 are proper. We define the composition of the corresponding correspondences by

$$[Z_{12}] \circ [Z_{23}] := p_{13,*}(p_{12}^*[Z_{12}] \cap p_{23}^*[Z_{23}]) \tag{5.1.1}$$

according to the diagram

$$\begin{tikzcd}
 & X_1 \times X_2 \times X_3 \arrow[dl, "p_{12}"'] \arrow[dd, "p_{13}"'] \arrow[dr, "p_{23}"] & \\
X_1 \times X_2 & & X_2 \times X_3 \\
 & X_1 \times X_3 &
\end{tikzcd}$$

which is well-defined since, in fact, the projection $p_{13}(p_{12}^{-1}Z_{12} \cap p_{23}^{-1}Z_{23}) \to X_3$ is proper (see [Nak, Section 8.3]).

6. Main construction

In this section we define the correspondences used in Nakajima's representation of the Heisenberg algebra and formulate the theorem which guarantees that it is so.

Let $n \geq i > 0$. Define $P[i]$ as a subset of $\coprod_n X^{[n-i]} \times X^{[n]}$ by

$$P[i] := \coprod_n \{(\mathcal{J}_1, \mathcal{J}_2, x) \in X^{[n-i]} \times X^{[n]} : \mathcal{J}_1 \supseteq \mathcal{J}_2, \operatorname{supp}(\mathcal{J}_1/\mathcal{J}_2) = \{x\} \text{ for some } x \in X\}.$$

For $i < 0$, we swap $\mathcal{J}_1$ and $\mathcal{J}_2$ in this definition. Now let $\alpha \in H_*^{BM}(X)$, $\beta \in H_*(X)$ and suppose that $i > 0$. Define the classes

$$\begin{aligned} P_\alpha[i] &= \varpi_*(\Pi_1^*\alpha \cap [P[i]]) \\ P_\beta[i] &= \varpi_*(\Pi_1^*\beta \cap [P[-i]]) \end{aligned} \tag{6.0.1}$$

according to the projections

$$\begin{tikzcd}
 & \coprod_n X^{[n-i]} \times X^{[n]} \times X \arrow[dl, "\Pi_1"'] \arrow[dr, "\varpi_i"] & \\
X & & \coprod_n X^{[n-i]} \times X^{[n]}.
\end{tikzcd}$$

These homology classes define the operators of the group $\bigoplus_{n=0}^{\infty} H_*(X^{[n]})$ that will be used in the main theorem.

Before we continue, we formulate and prove (with the help of other results, as explained in [Nak, Section 8.3]) a lemma that will be used later.

Lemma 6.1. $\dim_{\mathbb{C}} P[i] \cap (X^{[n-i]} \times X^{[n]}) = 2n - i + 1$.

Proof. Suppose $i > 0$. Fix $\mathcal{J}_1 \in X^{[n-i]}$. By Theorem 3.2 we know that $\dim X^{[n-i]} = 2(n-i)$. To find the dimension of $P[i]$ we must add to this number the dimension of the variety obtained by varying $\mathcal{J}_2 \in X^{[n]}$ satisfying $\mathcal{J}_1/\mathcal{J}_2 = \{x\}$ for some $x \in X$. The generic case is when $x \in X \setminus \operatorname{supp}(\mathcal{O}_X/\mathcal{J}_1)$ (this may be accepted under the explanation that since $\mathcal{O}_X/\mathcal{J}_1$ is the structure sheaf of a subvariety, its support has less dimension than that of X; a formal proof is provided in [Nak, Lemma 8.31]). In fact, the set of such $\mathcal{J}_2$ is $\pi^{-1}(S^i_{(i)}X)$, where π is the Hilbert-Chow morphism and $S^i_{(i)}(X) = \{i[x] \in S^i(X) : x \in X\}$. This notation is used in other parts of [Nak]; in our case just notice that $S^i_{(i)}(X)$ is the subset of the symmetric product $S^i(X)$ consisting of cycles of a single point with weight i. Finally, $\pi^{-1}(S^i_{(i)}(X))$ has complex dimension $i+1$ as a subvariety of the Hilbert scheme. $\square$

Now we state the main theorem, [Nak, Theorem 8.13]. Notice that we must add the assumption that the homology group of X is concentrated in even degrees. The theorem can be stated in a more general form without this assumption, but we shall not discuss it in this work.

Theorem 6.2 (Nakajima, Grojnowski)**.** *Suppose* $(deg\alpha)(deg\beta) \equiv 0(mod2)$ *for all* $\alpha, \beta \in H_*(X)$*. Then*

$$[P_\alpha[i], P_\beta[j]] = (-1)^{i-1} i\delta_{i,-j} \langle \alpha, \beta \rangle \, Id. \tag{6.2.1}$$

where $\langle \alpha, \beta \rangle = p_*(\alpha \cap \beta)$ *for* $p : X \to pt$ *the unique morphism from* X *to the point* pt*.*

According to our constructions in Section 2, this means that we have a representation of the Heisenberg algebra $\hat{\mathfrak{h}}$ associated to the graded vector space $H_*(X)$ on $\bigoplus_{n=0}^{\infty} H_*(X^{[n]})$.

Notice the factor $(-1)^{i-1}$ in Equation 6.2.1, which is not present in the Lie bracket equations of any of the variants of the Heisenberg algebra discussed earlier in this work. The reason for including this term in the main theorem is explained in [Nak, Theorem 9.15]. Moreover, this factor does not modify the Lie algebra structure: we can define a Lie algebra isomorphism from, say, the most basic version of the Heisenberg algebra with bracket given by Equation 2.0.1, to the Lie algebra $\bigoplus_{n\in\mathbb{Z}} \mathbb{C}\tilde{h}_n \oplus \mathbb{C}K$ with bracket

$$[\tilde{h}_m, \tilde{h}_m] = (-1)^{m-1} m\delta_{m,-n} K, \qquad K \text{ central},$$

by putting

$$h_m \mapsto \begin{cases} (-1)^{m-1}\tilde{h}_m & m < 0 \\ \tilde{h}_m & m \geq 0. \end{cases}$$

7. Worked example

In this section we do a worked example following [Nak] and [vid21, Video 10] to illustrate the main idea of the proof.

Fix $x \in X$ and let $\alpha = [X]$ be the BM class of X, and $\beta = [x]$ the class of the point x. Take a set of distinct n points $x_1, \ldots, x_n \in X \setminus \{x\}$ none of which is x. Identify them with the corresponding ideal $\mathcal{J}$. We refer interchangeably to the set of points and the ideal sheaf $\mathcal{J}$.

Let's show that

$$[P_\alpha[1], P_\beta[-1]][\mathcal{J}] = [\mathcal{J}].$$

We start by computing $P_\beta[-1][\mathcal{J}]$. For this consider

$$\begin{tikzcd} & X^{[n+1]} \times X^{[n]} \arrow[dl, "p_1"'] \arrow[dr, "p_2"] & \\ X^{[n+1]} & & X^{[n]}. \end{tikzcd}$$

By definition, $P_{[x]}[-1]$ is given by: pull back $\mathcal{J}$ via p_2, intersect with the class $\varpi_*(\Pi^*\alpha \cap [P[-1]]) = P_{[x]}[-1]$ and push forward down to $X^{[n+1]}$. This means that we push forward the cycle corresponding to

$$\{(\mathcal{J}, \mathcal{J}', x) : \mathcal{J} \subseteq \mathcal{J}', \mathrm{supp}(\mathcal{J}'/\mathcal{J}) = \{x\}\} \subset X^{[n]} \times X^{[n+1]} \times X.$$

The pushforward recovers the $X^{[n+1]}$ component, that is, the schemes $\mathcal{J}'$ in $X^{[n+1]}$ satisfying $\operatorname{supp}(\mathcal{J}'/\mathcal{J}) = \{x\}$. Since x is not a point in $\mathcal{J}$, the only possible choice is that $\mathcal{J}' = \{x\} \cup \mathcal{J}$, that is,

$$P_{[x]}[-1][\mathcal{J}] = [\{x\} \cup \mathcal{J}].$$

(see figure below).

On the other hand, the correspondence $P_{[X]}[1]$ starts with an ideal $\mathcal{J}' \in X^{[n+1]}$ intersects it with

$$\{(\mathcal{J}, \mathcal{J}', x) : \mathcal{J} \subseteq \mathcal{J}', \operatorname{supp}(\mathcal{J}'/\mathcal{J}) = \{y\}, \text{ for some } y \in X\}$$

and projects to $X^{[n]}$. We obtain $\sum_{y \in \mathcal{J}'} [\mathcal{J}' - \{y\}]$. Then it becomes clear that

$$[P_\alpha[1], P_\beta[-1]][\mathcal{J}] = [\mathcal{J}].$$

The following figure is taken directly from [Nak]:

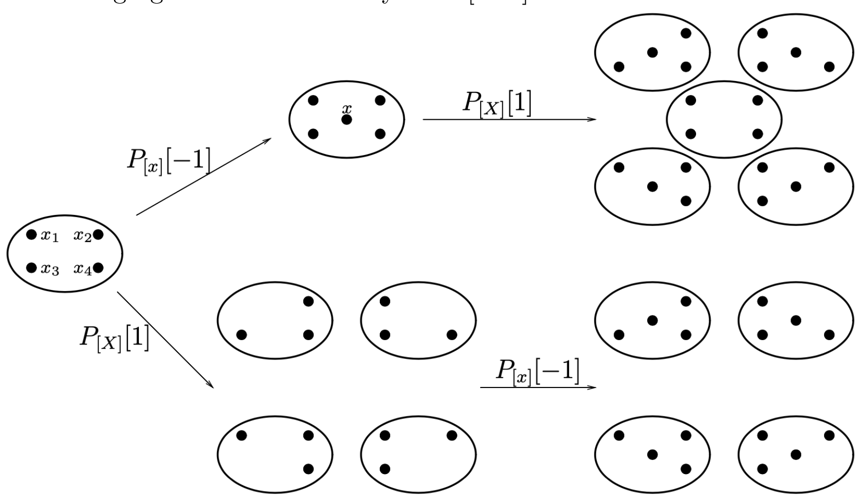

FIGURE 8.1. The proof of $\left[P_{[X]}[1], P_{[x]}[-1]\right][\mathcal{J}_1] = [\mathcal{J}_1]$

8. Proof of one case of the main theorem

The proof of Theorem 6.2 is split in two different cases:

(1) $i, j > 0$ or $i, j < 0$, and
(2) $i > 0$, $j < 0$ or $i < 0$, $j > 0$.

In this work we explain the proof of case (1).

Suppose $i, j > 0$ (the case $i, j < 0$ is analogous). Recalling our general definition for composition of correspondences (Equation 5.1.1) we see that the composition $P_\alpha[i]P_\beta[j]$ is given by

$$P_\alpha[i]P_\beta[j] = p_{13,*}(p_{12}^*[P[i]] \cap \Pi_1^*\alpha \cap p_{23}^*[P[j]] \cap \Pi_2^*\beta). \tag{8.0.1}$$

Consider the set

$$\begin{aligned} P[i,j] &:= p_{12}^{-1}(P[i]) \cap p_{23}^{-1}(P[j]) \cap (X^{[n-i-j]} \times X^{[n-j]} \times X^{[n]}) \\ &= \left\{(\mathcal{J}_1, \mathcal{J}_2, \mathcal{J}_3) : \mathcal{J}_1 \supset \mathcal{J}_2 \supset \mathcal{J}_3, {\scriptstyle \operatorname{supp}(\mathcal{J}_1/\mathcal{J}_2)=\{x\} \text{ for some } x\in X \atop \operatorname{supp}(\mathcal{J}_2/\mathcal{J}_3)=\{y\} \text{ for some } y\in X}\right\} \end{aligned} \tag{8.0.2}$$

which corresponds to the diagram

$$\begin{tikzcd}
 & X^{[n-i-j]} \times X^{[n-j]} \times X^{[n]} \arrow[dl, "p_{12}"'] \arrow[dr, "p_{23}"] \arrow[dd, "p_{13}"'] & \\
P[j] \subset X^{[n-j]} \times X^{[n]} & & X^{[n-i-j]} \times X^{[n-j]} \supset P[i] \\
 & X^{[n-i-j]} \times X^{[n]}. &
\end{tikzcd}$$

By definition, the correspondence $P[j]$ maps the homology of $X^{[n]}$ to the homology of $X^{[n-j]}$. In order to compose this with $P[i]$ we must regard $P[i]$ as a map from the homology of $X^{[n-j]}$ to the homology of $X^{[n-j-i]}$, which is perfectly valid since our correspondences are defined for all n. Observe that the middle factor $X^{[n-j]}$ in the triple product will become $X^{[n-i]}$ when we compute the composition in the opposite order.

To recover the correspondence as in Equation 8.0.1 from the set $P[i,j]$ it's natural to consider its fundamental class. We will do this in the following special way. Observe that $P[i,j]$ may be decomposed as a disjoint union of the set $\{x \neq y\}$ of "generic" cases in which x and y in Equation 8.0.2 are different, and the set $\{x = y\}$ in which they coincide. Let $\overline{P}[i,j] := \overline{P[i,j] \cap \{x \neq y\}}$. Then

$$p_{12}^*[P[i]] \cap p_{23}^*[P[j]] = [\overline{P}[i,j]] + \iota_* R$$

where $R \in H_*^{BM}(P[i,j] \cap \{x = y\})$ corresponds to the $\{x = y\}$ part of $P[i,j]$ and ι is the inclusion.

We can carry out the same construction for the composition $P_\beta[j]P_\alpha[i]$, obtaining

$$P_\alpha[j]P_\beta[i] = p_{13,*}(p_{12}^*[P[j]] \cap \Pi_2^*\alpha \cap p_{23}^*[P[i]] \cap \Pi_1^*\beta). \tag{8.0.3}$$

and

$$p_{12}^*[P[j]] \cap p_{23}^*[P[i]] = [\overline{P}[j,i]] + \iota'_* R'. \tag{8.0.4}$$

Observe that Equations 8.0.1 and 8.0.3 differ in the classes $\Pi_1^*\alpha$ against $\Pi_2^*\alpha$, and correspondingly with β. The following lemma is all we need to obtain the result:

Lemma 8.1. (1) *There exists an isomorphism*

$$P[i,j] \cap \{x \neq y\} \xrightarrow{\cong} P[j,i] \cap \{x \neq y\}, \tag{8.1.1}$$

which exchanges Π_1 *and* Π_2.

(2) *We have*

$$p_{13,*}(\Pi_1^*\alpha \cap \Pi_2^*\beta \cap \iota_* R) = 0, \qquad p_{13,*}(\Pi_1\beta \cap \Pi_2^*\alpha \cap \iota_* R') = 0. \tag{8.1.2}$$

Indeed, this lemma is all we need since

$$\begin{aligned}
P_\alpha[i]P_\beta[j] &= (-1)^{\deg\alpha} p_{13,*}((\overline{P}[i,j] + \iota_* R) \cap \Pi_1^*\alpha \cap \Pi_2^*\beta) \\
&= (-1)^{\deg\alpha\deg\beta} p_{13,*}((\overline{P}[j,i] + \iota_* R') \cap \Pi_1^*\beta \cap \Pi_2^*\alpha) \\
&= P_\beta[j]P_\alpha[i].
\end{aligned}$$

Proof of Lemma 8.1. (1) In this proof I assume that "isomorphism" means bijection, since $P[i,j]$ is considered as a set-theoretical intersection.

First notice that when we say that the isomorphism, say φ, "exchanges Π_1 and Π_2" we mean that $\Pi_1|_{X\times X} = \Pi_2|_{X\times X} \circ \varphi$ according to the diagram

$$\begin{tikzcd}
 & & X \\
P[i,j]\cap\{x\neq y\} \subset \arrow[ddd, "\varphi"'] & X^{[n-i-j]}\times X^{[n-j]}\times X^{[n]}\times X\times X \arrow[ur, "\Pi_1"] \arrow[dr, "\Pi_2"'] & \\
 & & X \\
 & & X \\
P[j,i]\cap\{x\neq y\} \subset & X^{[n-i-j]}\times X^{[n-i]}\times X^{[n]}\times X\times X \arrow[ur, "\Pi_1"] \arrow[dr, "\Pi_2"'] & \\
 & & X.
\end{tikzcd}$$

Now, the point of the isomorphism φ is that it must map an ideal in $X^{[n-j]}$ to an ideal in $X^{[n-i]}$. More precisely, we map $(\mathcal{J}_1, \mathcal{J}_2, \mathcal{J}_3) \in P[i,j] \cap \{x \neq y\}$ to $(\mathcal{J}_1, \mathcal{J}_2', \mathcal{J}_3) \in P[j,i] \cap \{x \neq y\}$. The defining relation is

$$\mathcal{J}_1/\mathcal{J}_2' = \mathcal{J}_2/\mathcal{J}_3, \qquad \mathcal{J}_2'/\mathcal{J}_3 = \mathcal{J}_1/\mathcal{J}_2. \tag{8.1.3}$$

The following figure (taken from [Nak]) shows how this correspondence works.

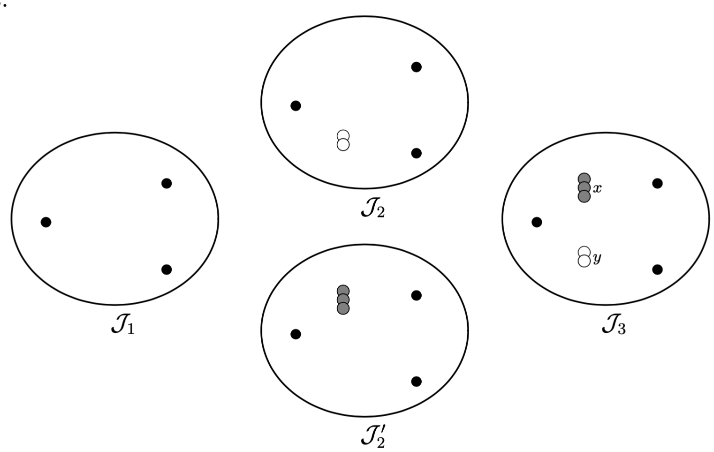

FIGURE 8.2. Correspondence between $\mathcal{J}_1 \supset \mathcal{J}_2 \supset \mathcal{J}_3$ and $\mathcal{J}_1 \supset \mathcal{J}_2' \supset \mathcal{J}_3$

The setting is the following. We have the ideals with contentions $\mathcal{J}_1 \supset \mathcal{J}_2, \mathcal{J}_2' \supset \mathcal{J}_3$. Passing to the supports, the inclusions are reversed, i.e the support of $\mathcal{J}_3$ contains the rest, and the support of $\mathcal{J}_1$ is contained in the rest.

To prove that this function is well-defined we need to show that for every $\mathcal{J}_2$ there is a unique $\mathcal{J}_2'$ satisfying Equation 8.1.3. Notice that by definition, $\mathrm{supp}\mathcal{J}_2/\mathcal{J}_3 = \{x\}$ for some $x \in X$ so that, as a scheme, $\{x\}$ has multiplicity j. We define $\mathcal{J}_2' = \mathcal{J}_1 \cup \{x\}$ where x has multiplicity j.

(Caveat: it looks to me that the choice of $\mathcal{J}_2'$ is not unique. Indeed, the space of schemes of length j supported in the point x was discussed in the proof of Lemma 6.1, where we used the fact that it has dimension $j-1$.)

Defining $\mathcal{J}_2'$ this way implies that

$$\mathcal{J}_1/\mathcal{J}_2' = \mathcal{J}_1/(\mathcal{J}_1 \cup \{x\}) = \{x\} = \mathcal{J}_2/\mathcal{J}_3.$$

Now, we also must have $\mathcal{J}_1/\mathcal{J}_2 = \{y\}$ for some $y \in X$ with multiplicity i. This implies that $\mathcal{J}_1/\mathcal{J}_3 = \{x\} \cup \{y\}$ with their multiplicities. Then

$$\mathcal{J}_2'/\mathcal{J}_3 = (\mathcal{J}_1 \cup \{x\})/\mathcal{J}_3 = \{y\} = \mathcal{J}_1/\mathcal{J}_2$$

as required. An inverse function defined similarly shows this is a bijection.

(2) Consider the commutative diagram

$$\begin{tikzcd} P[i,j] \cap \{x=y\} \arrow[r, "\Pi'"] \arrow[d, "\iota"'] & X \arrow[d, "\Delta"] \\ P[i,j] \arrow[r, "\Pi_1 \times \Pi_2"'] & X \times X \end{tikzcd}$$

where Δ is the diagonal embedding and Π' is any of the projections Π_1 or Π_2 (which coincide on $P[i,j] \cap \{x=y\}$). Then

$$\begin{aligned} \Pi_1^*\alpha \cap \Pi_2^*\beta \cap \iota_* R &= \iota_*(\iota^*(\Pi_1^*\alpha \cap \Pi_2^*\beta) \cap R) \\ &= \iota_*(\iota^*(\Pi_1 \times \Pi_2)^*(\alpha \times \beta) \cap R) \\ &= \iota_*({\Pi'}^*\Delta^*(\alpha \times \beta) \cap R) \\ &= \iota_*({\Pi'}^*(\alpha \cap \beta) \cap R), \end{aligned} \tag{8.1.4}$$

where the first equality is due to the projection formula (see Stacks Project 01E8) and we stick with Nakajima's notation $\alpha \times \beta$ for the tensor product $\alpha \otimes \beta$. We also have the commutative diagram

$$\begin{tikzcd} p_{13} \circ \iota : P[i,j] \cap \{x=y\} \arrow[rr, "\Pi'"] \arrow[d, "p_{13} \circ \iota"'] & & X \arrow[d, equal] \\ P[i+j] = \{(\mathcal{J}_1, \mathcal{J}_3) | \mathcal{J}_1 \supset \mathcal{J}_3, \mathrm{supp}(\mathcal{J}_1/\mathcal{J}_3) = \{x\}\} \arrow[rr, "\Pi''"'] & & X, \end{tikzcd}$$

where Π'' is the usual projection to X from the set $P[i+j]$.

By Equation 8.1.4 and the commutativity of the second diagram, we can express the class of interest as intersection of two classes: one pulled back directly from X, and the other one corresponding to R. That is, we have

$$p_{13,*}\iota_*({\Pi'}^*(\alpha \cap \beta) \cap R) = {\Pi''}^*(\alpha \cap \beta) \cap p_{13,*}\iota_*(R).$$

Now, by Lemma 6.1, we know that

$$\dim_{\mathbb{C}} P[i+j] = 2n - i - j + 1.$$

On the other hand, the "expected dimension" is $2n-i-j+2$. This obliges ι_*R to vanish.

(Caveat: I understand that the "expected dimension" means in [Nak] the dimension of $P[i,j]\cap\{x\neq y\}$, which is claimed to be $2n-i-j+2$. It is however not clear to me why this would imply that $p_{13,*}\iota_*R$ must vanish.) $\square$

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